

SAFETY DATA SHEET (GHS)

Sodium Carbonate (aqueous solution — caustic wash/neutralisation duty) | CAS: 497-19-8 | Formula: Na₂CO₃

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Sodium carbonate, CAS 497-19-8 (<https://echa.europa.eu/>)
- PubChem CID 10340 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/10340>
- ILO/NIOSH International Chemical Safety Card (ICSC) 1135 — Sodium carbonate
- Supplier SDS — Sodium carbonate solution (various concentrations), multiple suppliers
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Sodium Carbonate Solution (aqueous)

CAS Number: 497-19-8 (sodium carbonate); water CAS 7732-18-5

Chemical formula: Na₂CO₃ (solute); aqueous solution

Molecular weight: 105.99 g/mol (Na₂CO₃)

Synonyms: Soda ash (solid form); washing soda; disodium carbonate

Intended use: Caustic wash/neutralisation duty — used to neutralise residual acid (e.g. from the sulfuric acid-catalysed decomposition step) in process streams prior to further processing or discharge. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Eye Irrit. 2 (H319) — for solid/concentrated sodium carbonate and concentrated solutions; dilute working solutions may be below classification thresholds depending on concentration — confirm against the supplier SDS for the specific in-use concentration

Signal word: Warning

Hazard statements (H-codes): H319 (Causes serious eye irritation)

Precautionary statements (P-codes): P264, P280, P305+P351+P338, P337+P313

GHS pictograms: GHS07 (Irritant)

Other hazards: Sodium carbonate solutions are alkaline (mildly to moderately caustic depending on concentration) — concentrated solutions can cause skin/eye irritation and, at high concentrations, more severe effects; dust from solid soda ash (if handled in that form for make-up) is irritating to respiratory tract and eyes. Reacts with acids (including the sulfuric acid catalyst stream and any residual CHP/peroxide-bearing streams) releasing carbon dioxide gas — exothermic neutralisation; CO₂ release in confined spaces can contribute to localised oxygen displacement and pressure buildup if neutralisation occurs in a closed vessel.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Sodium carbonate: CAS 497-19-8 — concentration of the in-use aqueous solution should be confirmed against PPCL's neutralisation system design basis (typical industrial caustic wash solutions range from approximately 5-20% w/w Na₂CO₃, but this varies by application — CONFIRM ACTUAL CONCENTRATION)

Water: CAS 7732-18-5 — balance

SECTION 4 — FIRST AID MEASURES

Inhalation: If dust/mist is inhaled (e.g. during solid make-up operations), remove to fresh air. Obtain medical attention if respiratory irritation persists.

Skin contact: Wash with plenty of water. Remove contaminated clothing. Concentrated solutions may cause irritation with prolonged contact — obtain medical attention if irritation persists or skin damage occurs.

Eye contact: Flush immediately with copious water for at least 15 minutes, holding eyelids open. Obtain medical attention — alkaline materials can cause progressive eye damage if not adequately flushed (H319).

Ingestion: Rinse mouth with water and give water to drink if conscious. Do not induce vomiting. Obtain medical attention for significant ingestion — concentrated solutions can cause gastrointestinal irritation/burns.

Notes to physician: Treat as alkaline/caustic exposure proportional to concentration; for concentrated solution ingestion, assess for oesophageal/gastric irritation or burns

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Not combustible — use extinguishing media appropriate to surrounding fire

Unsuitable extinguishing media: None specific

Special hazards: None specific — non-flammable aqueous solution. Note: if this solution contacts strong acid streams in an uncontrolled manner (e.g. during a process upset), the resulting neutralisation reaction releases CO₂ gas and heat — consider this in scenarios involving co-located acid (sulfuric acid) and caustic systems

Special protective equipment for firefighters: Standard firefighting PPE; no special requirements beyond general chemical exposure precautions

Hazardous combustion products: None — aqueous inorganic solution is non-combustible

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Wear appropriate PPE (Section 8) to avoid skin/eye contact with concentrated solution. Be aware of slip hazard on wet/alkaline surfaces.

Environmental precautions: Prevent large releases to drains/watercourses — elevated pH can affect aquatic life and receiving water treatment processes; neutralise before discharge if outside permitted pH range

Containment and cleanup: Contain spill; absorb with inert absorbent or flush to appropriate collection/neutralisation system. Wash residual area with water.

Reference to other sections: See Section 7 (Handling/Storage) and Section 10 (reactivity with acids)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Avoid skin/eye contact with concentrated solution; use appropriate PPE for transfer/dosing operations. If solid soda ash is used for make-up, avoid dust generation/inhalation.

Storage conditions: Store in suitable corrosion-resistant tanks (carbon steel is generally suitable for sodium carbonate solutions, unlike strong acids); keep containers closed when not in use

Temperature limits: Store at ambient temperature; monitor for crystallisation at low temperatures if solution concentration is near saturation (sodium carbonate solubility decreases at low temperature for some hydrate forms)

Pressure: Atmospheric storage

Segregation requirements: Segregate from strong acids (notably the sulfuric acid catalyst, CAS 7664-93-9) to prevent uncontrolled neutralisation reactions (CO₂ release, heat, potential for vessel overpressure if mixed in a closed vessel); segregate from CHP/peroxide-containing streams unless the neutralisation/wash step is the intended, controlled point of contact

Special requirements: Where this solution is used to neutralise acid-bearing process streams, ensure adequate venting/pressure relief on the neutralisation vessel to accommodate CO₂ evolution

Specific end-use recommendations: Confirm the specific neutralisation step(s) in the Train II process where this solution is used and verify that controlled, metered addition (rather than bulk/uncontrolled mixing) is designed into the process to manage the exothermic CO₂-releasing neutralisation reaction

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): No specific ACGIH TLV or OSHA PEL established for sodium carbonate; general nuisance dust limits may apply for solid handling (e.g. OSHA PEL for particulates not otherwise regulated, ~15 mg/m³ total dust / 5 mg/m³ respirable). Confirm Thailand OEL if applicable.

TWA: Not specifically established (see nuisance dust note above for solid handling)

STEL: Not established

Ceiling: Not established

Skin notation: Not assigned

IARC/NTP/OSHA carcinogen status: Not classified as a carcinogen

Engineering controls: General ventilation adequate for solution handling; local exhaust/dust control if solid soda ash make-up is performed

PPE — Respiratory: Not normally required for solution handling; dust mask/respirator for solid make-up operations with dust generation

PPE — Gloves: Chemical-resistant gloves (e.g. nitrile, PVC, rubber) for handling concentrated solution

PPE — Eyes: Chemical splash goggles for tasks with splash potential (H319)

PPE — Body: Standard work clothing; chemical-resistant apron for bulk transfer/dosing tasks

Biological monitoring: Not applicable

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Clear, colourless to slightly turbid aqueous solution (solid Na₂CO₃ is a white crystalline powder/granules)

Odour: Odourless

Odour threshold: Not applicable

pH: Strongly alkaline — pH approximately 11-12 for typical aqueous solutions (concentration-dependent)

Melting point / Freezing point: Solid Na₂CO₃ melts/decomposes at ~851°C (not relevant to aqueous solution; solution freezing point depressed below 0°C depending on concentration)

Boiling point (at 1 atm): Solution boiling point close to that of water, slightly elevated depending on concentration (~100-105°C for dilute-to-moderate solutions)

Flash point: Not applicable — aqueous inorganic solution, non-flammable

Evaporation rate: Similar to water

Flammability (solid/gas): Not flammable

Flammability limits in air: Not applicable

Vapour pressure (at 20°C): Approximately that of water, slightly reduced by dissolved solids

Vapour density (air = 1): Not applicable (aqueous solution; vapour is essentially water vapour)

Relative density (water = 1): ~1.05-1.20 depending on concentration (solid Na₂CO₃ has density ~2.54 g/cm³)

Solubility (water): Solute is highly soluble in water (~215 g/L at 20°C for anhydrous Na₂CO₃) — solution is by definition fully miscible/homogeneous

Partition coefficient (log K_{ow}): Not applicable (inorganic ionic solute)

Auto-ignition temperature: Not applicable

Decomposition temperature: Solid Na₂CO₃ is thermally stable up to its melting point (~851°C); not relevant to aqueous solution under process conditions

Viscosity: Slightly higher than water, concentration-dependent

Explosive properties: Not classified as explosive

Oxidising properties: Not an oxidiser

Molecular weight: 105.99 g/mol (Na₂CO₃)

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Stable under normal conditions; reacts with acids to release carbon dioxide gas (effervescence) and heat — this is the basis of its intended neutralisation function

Chemical stability: Stable under recommended storage conditions

Possibility of hazardous reactions: Reacts with acids (e.g. sulfuric acid, CAS 7664-93-9) to produce CO₂ gas and heat — controlled/metered addition is intended, but uncontrolled mixing of concentrated acid and carbonate solution can produce vigorous effervescence/foaming and a rapid CO₂ release that could pressurise a closed vessel; reacts with certain metal salts to form precipitates (relevant to scaling/fouling considerations in process equipment, not a safety hazard per se)

Conditions to avoid: Uncontrolled/rapid mixing with concentrated acids in closed vessels without adequate venting

Incompatible materials: Strong acids (controlled reaction is intentional; uncontrolled mixing is the hazard); some metal salts (precipitation/scaling)

Hazardous decomposition products: Reaction with acids produces CO₂ gas — monitor for localised oxygen displacement/pressure effects in confined neutralisation vessels

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Skin/eye contact (irritation, H319), inhalation of dust (solid form), ingestion

Acute toxicity — Oral LD₅₀: ~4,090 mg/kg, rat (low acute oral toxicity)

Acute toxicity — Dermal LD₅₀: Not well characterised; low acute dermal toxicity expected for typical solution concentrations

Acute toxicity — Inhalation LC₅₀: Not established; dust from solid form is irritating to respiratory tract

Skin corrosion/irritation: Concentrated solutions/solid may cause irritation; typical dilute working solutions less severe — confirm against the specific in-use concentration

Serious eye damage/irritation: Causes serious eye irritation (H319)

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: Not classified

Carcinogenicity: Not classified as a carcinogen

Reproductive toxicity: Not classified

STOT (single exposure): Not classified

STOT (repeated exposure): Not classified

Aspiration hazard: Not applicable

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC₅₀/EC₅₀): Low intrinsic toxicity; ecological effects of releases are primarily related to pH elevation rather than direct toxicity — fish/invertebrate LC₅₀/EC₅₀ values are generally high (low toxicity) but pH effects on receiving waters should be assessed

Persistence and degradability: Not applicable (inorganic salt; dissociates completely in water)

Bioaccumulative potential: Not applicable

Mobility in soil: Highly mobile (soluble inorganic salt)

Other adverse effects: Discharge of alkaline solutions can raise receiving water pH — manage per site effluent discharge permit (pH limits)

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Neutralise to within permitted pH range before discharge per site effluent permit; dispose of any concentrated waste solution via appropriate industrial wastewater treatment or licensed waste contractor

Contaminated packaging: Rinse with water; dispose of per local regulations

Regulatory reference: Site effluent discharge permit (pH limits); Thailand Hazardous Substance Act B.E. 2535 (verify applicability)

SECTION 14 — TRANSPORT INFORMATION

UN Number: Not regulated as a dangerous good for transport (sodium carbonate solution at typical concentrations is generally not assigned a UN number); confirm for the specific concentration/product used on-site

Proper shipping name: Not applicable / Not regulated (confirm for specific product)

Transport hazard class(es): Not classified as a transport hazard at typical concentrations

Packing group: Not applicable

Environmental hazards (marine pollutant): Not classified as a marine pollutant

Special precautions for user: None specific beyond standard good practice for bulk liquid transfer

Bulk transport (MARPOL/IBC): Not typically subject to IBC Code special provisions at typical concentrations

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 497-19-8). NOTE: in-use solution concentration should be confirmed against PPCL's neutralisation system design basis for precise Section 9 property values and Section 2 classification at the actual working concentration.

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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